

HTML NOTES

HTML = HyperText Markup Language

HTML is used to create and structure the content of web pages. It defines headings, paragraphs, links, images, tables, forms and other page elements.

1. Basic HTML Structure

Key points

- HTML documents normally contain `<!DOCTYPE html>`, `<html>`, `<head>` and `<body>`.
- The `<head>` contains metadata, title, CSS links and scripts.
- The `<body>` contains the visible page content.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>My Page</title>
</head>
<body>
<h1>Hello World</h1>
</body>
</html>
```

2. HTML Tags and Elements

Important concepts

- Tag: A markup instruction such as `<p>`.
- Element: Usually consists of an opening tag, content and closing tag.
- Attribute: Provides additional information, e.g. href, src, alt, id, class.

Common tags

- `<h1>`–`<h6>` : Headings
- `<p>` : Paragraph
- `<br>` : Line break
- `<hr>` : Horizontal line
- `<a>` : Hyperlink
- `<img>` : Image

- <div> : Block container
- : Inline container

3. Text Formatting

Tags

- or : Bold / strong importance
- <i> or : Italic / emphasis
- <u> : Underline
- <mark> : Highlighted text
- <small> : Smaller text
- <sub> : Subscript
- <sup> : Superscript

4. Lists in HTML

Types

- Ordered list: with items.
- Unordered list: with items.
- Description list: <dl>, <dt>, <dd>.

Example

```
<ol>
<li>HTML</li>
<li>CSS</li>
</ol>

<ul>
<li>C++</li>
<li>Python</li>
</ul>
```

5. Hyperlinks

Syntax

```
<a href="https://example.com">Visit Website</a>
```

Important attributes

- href: Destination URL.
- target="_blank": Opens the link in a new browsing context.
- title: Gives additional information about the link.

6. Images



Syntax

```

```



Important attributes

- src: Image location.
- alt: Alternative text.
- width / height: Image dimensions.

7. HTML Tables

Main tags

- <table> : Creates table
- <tr> : Table row
- <th> : Header cell
- <td> : Data cell
- rowspan : Merges cells vertically.
- colspan : Merges cells horizontally.

Example

```
<table border="1">
<tr><th>Name</th><th>Marks</th></tr>
<tr><td>Amit</td><td>85</td></tr>
</table>
```

8. HTML Forms

Important tags

- <form> : Defines a form.
- <input> : Accepts user input.
- <label> : Describes an input.
- <textarea> : Multi-line text.
- <select> and <option> : Drop-down list.
- <button> : Button.

Common input types

- text, password, email, number, radio, checkbox, date, file, submit, reset

9. Multimedia

Tags

- <audio> : Embeds audio.
- <video> : Embeds video.
- <source> : Specifies media source.



- `<iframe>` : Embeds another webpage or external content.

10. HTML5 Semantic Elements

Important tags

- `<header>` : Introductory/header area
- `<nav>` : Navigation links
- `<main>` : Main content
- `<section>` : Thematic section
- `<article>` : Independent content
- `<aside>` : Side content
- `<footer>` : Footer information

Advantages

- Improves page structure and readability.
- Helps accessibility and search-engine understanding.
- Makes web pages easier to maintain.

11. HTML5 Image Map

Concept

- An image map makes different areas of one image clickable.
- `<map>` defines the map and `<area>` defines clickable regions.
- Common shapes include rect, circle and poly.

Example

```

<map name="campusmap">
<area shape="rect" coords="0,0,120,100" href="library.html">
</map>
```

12. HTML Entities

Examples

- `<` → `<`
- `>` → `>`
- `&` → `&`
- ` ` → non-breaking space
- `©` → ©

13. HTML vs CSS vs JavaScript

Quick comparison

- HTML: Structure and content.

- CSS: Presentation, layout and styling.
- JavaScript: Behaviour and interactivity.

14. Important Exam Points

Remember

- HTML is a markup language, not a programming language.
- HTML5 is the modern HTML standard.
- Attributes are written inside the opening tag.
- Use semantic elements when their meaning matches the content.
- Always provide meaningful alt text for important images.
- Tables are intended for tabular data, not general page layout.

Quick Revision

Topic	Remember
HTML	Structure of a webpage
CSS	Style and layout
JavaScript	Behaviour and interactivity
<a>	Link
	Image
<table>	Table
<form>	User input
<map> + <area>	Clickable image regions
Semantic tags	Meaningful page structure